

SymPy Bug Card

Bug 21: solveset returns 1 for unsatisfiable principal-log equation

Task. Solve the principal-log equation over the complex numbers.

$$\log(x) = 2\pi i, \quad x \in \mathbb{C}.$$

SymPy's answer.

$$\text{solveset}(\log(x) = 2\pi i, x, \mathbb{C}) = \{1\}.$$

Correct answer.

$$\emptyset.$$

Explanation. Substituting SymPy's returned value gives $\log(1) - 2\pi i = -2\pi i \neq 0$. Since $\log$ is the principal logarithm, its imaginary part lies in $(-\pi, \pi]$, so $2\pi i$ is outside its image.

Likely root cause. In `sympy/solvers/solveset.py`, `_invert_complex` appears to use `exp` as a global inverse for `log` without checking the principal-branch range, and the resulting finite candidate is not rejected by a residual check. This is a new instance of a known failure mode.

Artifact (GitHub). [bug-report/bug-021-solveset-returns-1-for-unsatisfiable-principal-log-equation/](#)